Motivated Affective Polarization: Recall of relative partisan generosity

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Abstract

Affective polarization poses significant social and political concerns. Interventions can reduce it in the short term, but effects often decay over time. This paper tests one explanation: individuals are motivated to sustain polarized beliefs about partisan groups even after receiving contrary information, and do so through memory. In online experiments, partisans observe signals of partisan generosity, with recall measured immediately or after one week. I find that partisans systematically misremember unfavorable information about their ingroup, with distortion increasing over time. These memory distortions have downstream consequences for polarized beliefs and intergroup discrimination. The results identify motivated memory as a mechanism behind the persistence of affective polarization and highlight the long-term limitations of interventions designed to reduce it.

1 Introduction

Affective polarization, or animosity between political parties, poses significant challenges to social cohesion and democratic governance. One of its most striking manifestations is the systematic attribution of negative traits to political opponents while ascribing positive traits to one's own party members. Survey research consistently shows that both Democrats and Republicans characterize members of the opposing party as more "selfish" and less "generous" than their co-partisans (Iyengar et al. 2019; Garrett et al. 2014; Levendusky 2018). These perceptions are mutually inconsistent, and they are contradicted by behavioral evidence. Fowler & Kam (2007) find that in standard dictator games, Democrats and Republicans allocate similar amounts to anonymous recipients.

Several factors could explain polarized inter-party perceptions. Increasingly homogeneous social networks (Gimpel & Hui 2015) and the proliferation of partisan news media (Lelkes et al. 2017) may provide insufficient or inaccurate information about members of the opposing party. However, these explanations do not account for why perceptions persist even when people encounter contrary information, as documented in interventions in which initial reductions in affective polarization decay over time (Santoro & Broockman 2022; Voelkel et al. 2024; Holliday et al. 2025). This raises a fundamental question: What psychological mechanisms sustain affective polarization even after encountering contrary information?

This paper investigates whether motivated beliefs, supported by biased memory, serve as a key mechanism linking partisan identity to persistent belief polarization. According to social identity theory (Tajfel & Turner 1979), individuals are motivated to maintain a positive social identity, initiating thoughts, feelings, and behaviors that positively differentiate the ingroup from the outgroup. When a positive view of their identity is challenged, individuals may attempt to maintain a positive group image because of the value attached to holding these beliefs, whether as a direct or instrumental source of utility (Bénabou & Tirole 2002, 2016). Consequently, they may correctly recall information portraying their group favorably while misremembering information portraying their group unfavorably, to sustain a positive view of their identity.

I conduct online experiments to investigate motivated memory about relative partisan generosity. In Experiment 1, participants identifying as Democrats or Republicans played a one-shot dictator game with an anonymous partner, providing a behavioral measure of generosity free from reputation and reciprocity concerns. In Experiment 2, a different set of subjects observed the outcomes of seven dictator game choice pairs, each comparing a Democrat's choice with a Republican's. Crucially, I manipulate the time between signal observation and recall, comparing immediate recall with recall after one week. Subjects were then incentivized to recall the number of pairs in which their ingroup member was more generous than the outgroup member.

My findings provide novel evidence that motivated memory serves as a mechanism through which partisan identity influences the formation of beliefs regarding group traits. I empirically confirm the belief-behavior discrepancy: approximately 76 percent of partisans expect their ingroup to be relatively more generous than the outgroup despite no significant differences in actual dictator game behavior between Democrats and Republicans. More importantly, subjects systematically recalled information about relative partisan generosity in favor of their group, and this recall bias was more pronounced after one week. Exploiting exogenous variation in signal valence, I show that this bias is driven primarily by unfavorable signals—information that portrays the ingroup negatively is more often misremembered and becomes increasingly distorted over time.

These memory distortions have direct consequences for forming beliefs about relative partisan generosity, with a strong positive relationship between recalled signals and posterior beliefs. Using the experimental timing manipulation as an instrument, I establish a causal interpretation for the effect of the recall bias on beliefs. Subjects who received unfavorable signals initially formed beliefs that did not indicate affective polarization on average, but one week later exhibited affective polarization. Finally, these polarized beliefs are related to intergroup behavior: subjects exhibit a greater willingness to pay to be matched with ingroup rather than outgroup dictators and are less likely to punish ingroup members for selfish behavior.

These findings contribute to several streams of social science literature. First, I contribute to the understanding of the persistence of affective polarization and the limitations of interventions designed to reduce it. While evidence shows that intergroup contact and correcting misperceptions can reduce affective polarization (Levendusky & Stecula 2021; Ahler & Sood 2018; Voelkel et al. 2023), effects may decay over time (Santoro & Broockman 2022; Voelkel et al. 2024; Holliday et al. 2025). My results suggest that a novel mechanism, namely motivated memory, is a channel through which interventions may fail to create durable change. Moreover, I identify a potential solution: participants who immediately processed signals into beliefs after exposure did not exhibit further memory distortion one week later, suggesting that interventions incorporating immediate belief consolidation such as reflection exercises or comprehension quizzes may impose constraints that mitigate motivated memory.

Next, I contribute to the literature on the behavioral consequences of affective polarization. Previous work has documented that partisanship spills over into economic decisions (Panagopoulos et al. 2020; McConnell et al. 2018) and political behavior (Pierson & Schickler 2020), but has relied primarily on correlational evidence. I provide, to my knowledge, novel evidence on the causal effect of polarized partisan beliefs on costly discrimination against outgroup members and weakened accountability within one's own group.

The paper also contributes to an empirical literature on motivated memory. Prior studies on motivated memory in economic settings have documented self-serving recall errors for personal behavior (Zimmermann 2020; Carlson et al. 2020; Huffman et al. 2022; Saucet & Villeval 2019; see Amelio & Zimmermann 2023 for a review), but have not focused on memory distortion of others' behavior in group contexts.[1] The closest antecedent is Li (2013), who documents asymmetric recall of others' kind and unkind acts, though in interactions where the subject was personally affected and self-serving motives remain available. In contrast, subjects in my experiment recall the choices of strangers with whom they never interacted, so any distortion serves the group's image rather than their own. I show that motivated memory operates not only to protect individual self-image but also to maintain positive group identity, revealing memory as a mechanism that sustains biased intergroup beliefs even in the face of contrary evidence.

Finally, I contribute to the literature on identity and belief formation in economics. While extensive research documents the influence of social identity on economic behavior starting with Akerlof & Kranton (2000)[2], its role in belief formation has received less attention until more recently (Kahan 2013; Bauer et al. 2023; Dekel & Shayo 2024). Recent work by Bauer et al. (2023) provides important insights into how group identity affects the belief formation process. My study complements this research in two key ways. First, while they focus on how partisans selectively seek and interpret information, I examine a previously understudied stage of belief formation: memory and recall of information. Second, they investigate ideological polarization around policy issues, while I focus on affective polarization regarding group traits. Together, these complementary approaches provide a more comprehensive picture of how partisan identity shapes the belief formation process. Their work demonstrates that identity affects what information people seek and how they interpret it initially, while my findings show that identity continues to influence beliefs through motivated memory even after information has been acquired and processed. This suggests that the influence of partisan identity on beliefs operates through multiple reinforcing mechanisms, shedding light on why partisan beliefs can be so prevalent and persistent.

The remainder of this paper is organized as follows. The experimental design is described in Section 2. The results are presented in Section 3. Section 4 concludes.

[1]Studies in the social psychology literature have explored the role of social identity in memory of information about others (Howard & Rothbart 1980; Mackie & Worth 1989; Park & Rothbart 1982; Park & Judd 1990; Sherman et al. 1998; Schaller & Maass 1989; Gramzow et al. 2001; Hechler et al. 2016). The most notable difference between my study and those in social psychology is the use of financial incentives. These studies often rely on hypothetical or non-incentivized behaviors rather than incentivized actions in controlled settings.

[2]See Charness & Chen (2020) for a review.

2 Experimental Design

2.1 Overview

To investigate the presence of motivated memory about relative group generosity and its implications, I use an experimental setup in which subjects (Democrats and Republicans) receive (i) exogenously varying signals about relative group generosity, and (ii) a manipulation of the time span between observing the signal and recall, which isolates memory effects on recall, belief formation, and intergroup behavior.

I conduct two experiments on the online platform *Prolific* using oTree software (Chen et al. 2016) with participants located in the United States. Experimental currency units were used with 100 tokens equal to $1, and one in ten participants received bonus payments based on their choices.

2.2 Experiment 1: Political group dictator game choices

2.2.1 Design

Experiment 1 generates the choice data that forms the signal over which beliefs are formed in the main experiment. To obtain a measure of generosity free from reciprocity and reputation concerns, participants played the dictator game (Forsythe et al. 1994). The game was a one-shot, anonymous dictator game with role uncertainty and a giving multiplier of three. This experiment provides the empirical foundation for Experiment 2 by generating comparable dictator game choice pairs between Democrats and Republicans such that relative generosity between the two participants can be determined.

2.2.2 Procedure

Participants were randomly matched into pairs and simultaneously made allocation decisions over 100 tokens between themselves and an anonymous partner. Both players' choices were equally likely to be implemented. After the allocation decision, participants completed a questionnaire collecting demographic information and political affiliation. A total of 193 participants completed Experiment 1 in April 2024. Participants were recruited based on the criteria that they were located in the United States and reported a U.S. Political Affiliation of Democrat or Republican. All participants received $1 for finishing the study, in addition to the potential bonus payment.[3]

[3]See Appendix Table A1 for summary statistics and balance for Experiment 1.

2.3 Experiment 2: Motivated memory and belief formation

2.3.1 Design

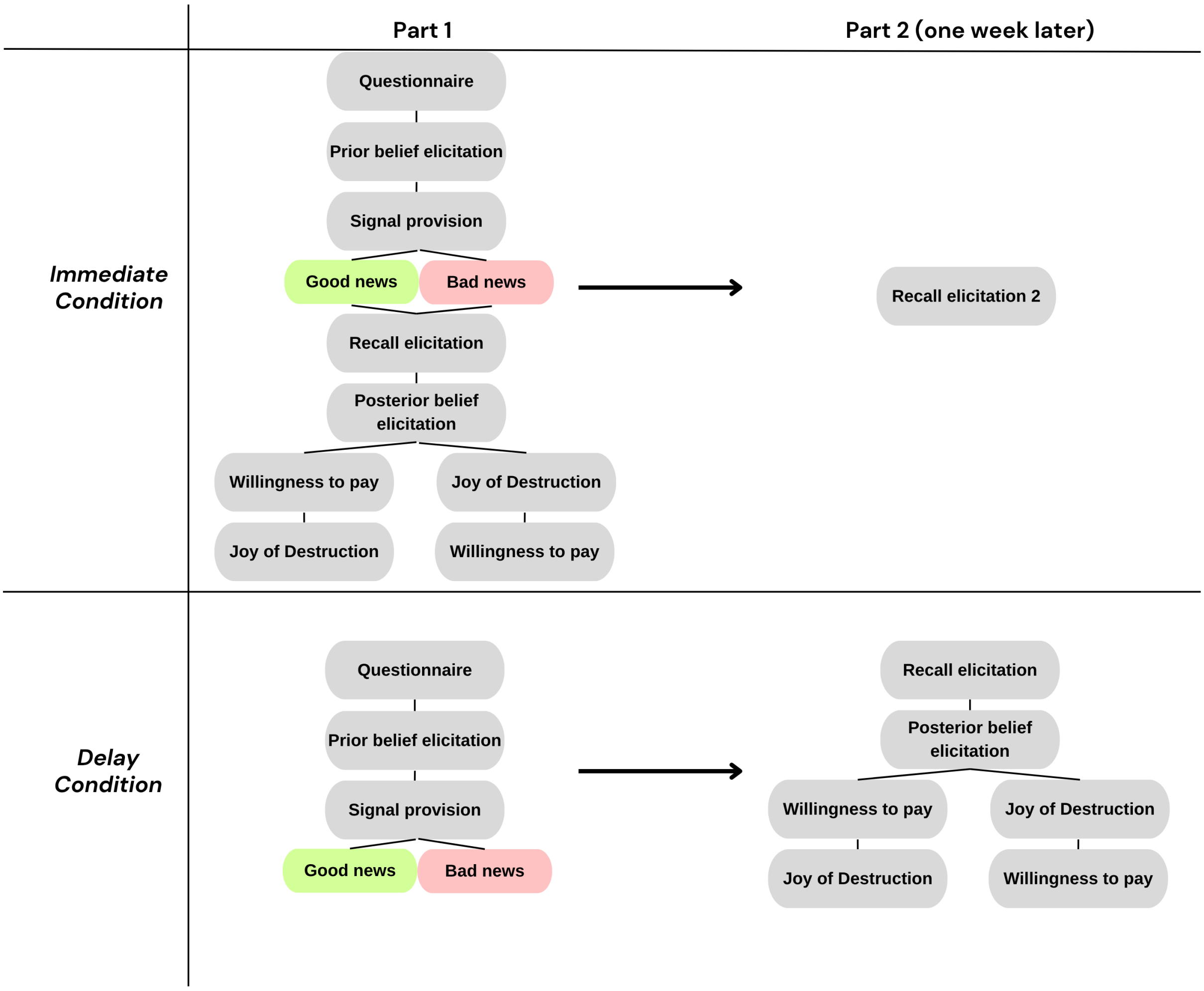
Experiment 2 was preregistered on AsPredicted.org (#200626) before data collection. The experiment investigates how identity affects memory about relative group generosity, as well as downstream effects on belief formation and intergroup behavior through a 2×4 factorial design embedded in a two-part, week-long study. The experimental design manipulates two types of variation to identify motivated memory effects.

Signal Valence: Participants observe exactly seven pairs of Democrat-Republican dictator game choice pairs from Experiment 1, with the number of pairs favorable to their group varying across four conditions: 1, 3, 4, or 6 favorable pairs out of 7. This creates a differential need for motivated reasoning, since participants receiving an unfavorable signal (1/7) about the relative generosity of their group face stronger psychological incentives to forget or misremember compared to those receiving a favorable signal (6/7). The signal valence is randomly assigned but generated from actual Experiment 1 choice data.

Recall Timing: The timing between observing the signal and recalling the signal varies across two conditions: *immediate* (recall directly after observation in Part 1) and *delay* (recall one week later in Part 2). This manipulation isolates the effect of memory decay on recall and subsequent tasks which are described in the next section. To avoid selection effects based on treatment conditions, all participants were recruited into the same study, which explicitly required returning for Part 2 to receive the financial reward. Returning subjects in the *immediate* condition also repeat the recall task in Part 2, providing a within-subject comparison of recall with and without delay.

The main structure of the experiment is as follows: the prior belief elicitation establishes initial beliefs about relative group generosity; the signal provision delivers the experimental signal of relative group generosity; the recall elicitation measures recall of the observed signal and represents the primary outcome of interest; and the posterior belief elicitation and intergroup choices capture downstream consequences for beliefs and behavior. Further details on each of these components are described below.

Figure 1: Design Overview



2.3.2 Procedure

Part 1: The experiment began with a questionnaire about political affiliation, with Independents classified by partisan leaning. Participants then read a description of the Experiment 1 dictator game and were required to pass comprehension questions to ensure understanding of the choice environment. Those with two incorrect attempts were excluded from the study.

Prior beliefs were elicited using an incentivized interval method over 99 randomly generated Democrat-Republican choice pairs from Experiment 1. Participants guessed how many pairs resulted in the outgroup as the 'lesser giver', with a 100 token bonus for accuracy within two pairs. This provides individual-level baseline measures of affective polarization and enables tracking belief dynamics.

Participants next observed the signal, which is a subset of seven choice pairs drawn from the 99 randomly generated choice pairs, according to their treatment assignment. A table displayed each pair with the political affiliation of the lesser giver clearly indicated. To ensure attention, participants were required to click the correct political affiliation for each pair before proceeding (see appendix for screenshot). Subjects in the *delay* condition were done with Part 1 at this point, and were asked to return one week later to complete Part 2.

Right after observing the signal, subjects in the *immediate condition* completed the recall task, which asked participants to recall both the total number of pairs observed (7) and the number where the outgroup was the lesser giver. The latter serves as the main outcome measure and is incentivized with 100 tokens for accurate recall. The possible options were "0" through "7," and "I don't recall."[4]

Posterior beliefs were elicited next, giving subjects another chance to guess the number of pairs out of 99 in which the outgroup subject was the 'lesser giver', with the same incentive as the prior elicitation. This allows measuring belief dynamics over time and investigating the relationship between recall and belief formation at the individual level.

The intergroup choice tasks measure behavioral consequences through two incentivized decisions with economic consequences: (i) willingness-to-pay to be matched with an ingroup rather than an outgroup dictator, and (ii) a modified joy of destruction game allowing participants to reduce payoffs of randomly selected ingroup or outgroup members by up to \$1 at no cost to themselves. The order of these two tasks was randomized.

In the WTP choice, by default, subjects received an allocation made by an outgroup dictator, but their willingness to pay to be matched with an ingroup dictator instead was elicited. In a multiple price list (MPL) with positive and negative prices, subjects indicated at which row they prefer to switch from being matched with an outgroup dictator to being matched with an ingroup dictator. Subsequently, a single row was randomly drawn and implemented. If the subject preferred to match with an outgroup for the implemented row, they received an allocation from a randomly chosen outgroup dictator from Experiment 1, and if they preferred to match with the ingroup for the implemented row, they received an allocation from a randomly chosen ingroup dictator from Experiment 1, with a corresponding payoff adjustment. For this choice, beliefs about relative group generosity are both relevant, since subjects should be willing to pay a premium if they expect to receive more from an ingroup than from an outgroup dictator. At the same time, if Democratic and Republican dictators give equal amounts to their partners on average, a positive WTP represents an expected welfare loss.

Next, in a modified version of the joy-of-destruction game by Abbink & Sadrieh (2009), subjects were given an option to destroy up to 100 tokens from the payment of a random ingroup member or outgroup member at no cost to themselves. They could also choose to take no action. In the original version of the game, subjects were anonymously paired up with each other, and fairness motives are absent. In this study, the choice differs in two key ways. First, subjects were not paired up with each other anonymously.

[4] "I don't recall" was included to ensure a more reliable measure of recall, allowing subjects who forgot the signal to opt out of the task. I address selection bias concerns in Appendix section A.3.

Rather, they selected the partisan identity of the individual whose payoff they would like to reduce, and they were subsequently matched with a random subject in the study identifying with that partisan group. Second, it occurred after subjects observed how Democrats and Republicans behaved in a dictator game and formed beliefs about relative group generosity. Consequently, this choice gave subjects an opportunity to punish a group that they perceived as selfish. See Appendix A.7 for screenshots of the main tasks in the study.

Part 2: One week after Part 1, all participants received invitations to complete Part 2.[5] After brief reminders about what happened in Part 1, returning participants in the *delay* condition completed the recall elicitation and subsequent tasks using identical procedures as the *immediate* condition. Returning participants in the *immediate* condition only completed the recall elicitation, for a second time.

A total of 440 participants completed Part 1 of Experiment 2 on December 3rd, 2024, and 390 of them returned and completed Part 2 between December 10th and 12th, 2024, with an overall attrition rate of 11.36%. There is no evidence of differential attrition across observable characteristics or experimental manipulations.[6] Participants were recruited using the same criteria as in Experiment 1, but with the additional restriction of having completed 20 or more submissions on Prolific. The aim was to recruit the same number of participants in each information condition, but with a larger proportion assigned to the *delay* condition (60%) than to the *immediate* condition (40%), so that the ratio of sample sizes was equal to the expected ratio of the standard deviations of outcomes in the conditions. After randomization, prior beliefs were imbalanced, with subjects in the *immediate* condition exhibiting more ingroup-favoring priors on average than subjects in the *delay* condition. Ingroup-favoring priors are not negatively associated with ingroup-favoring recall, and if anything, show a weakly positive relationship. Therefore, the imbalance in priors would, if anything, bias against detecting stronger ingroup-favoring recall after one week in the *delay* condition relative to the *immediate* condition. Nevertheless, all regression analyses include specifications that control for participants' prior beliefs.[7] All participants received $2.20 for finishing both parts of the study, in addition to the potential bonus payment.

[5] Part 2 of the experiment began exactly one week after Part 1 began, but remained open for approximately 50 hours to allow more participants to return and minimize the attrition rate. The maximum delay experienced by a subject between Part 1 and Part 2 was 9 days and 1 hour.

[6] See Appendix Table A5 for details on attrition.

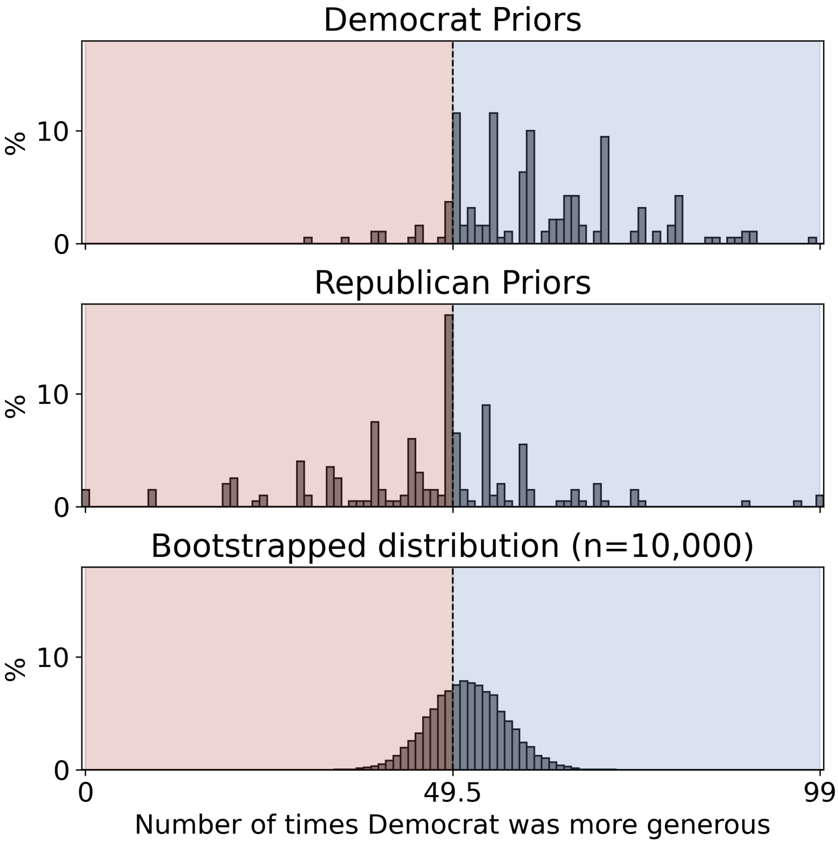
[7] See Appendix Table A2, A3, and A4 for summary statistics and balance for the final sample.

3 Results

3.1 Empirical validation of core belief discrepancy

This section empirically tests the core discrepancy that motivated the present research: partisans perceive members of the opposing party as less generous than members of their own party, yet experimental evidence suggests there are similar levels of generosity between Democrats and Republicans. Subjects in Experiment 1 gave approximately 35% of their endowment to an anonymous partner in a dictator game, with no evidence of a partisan difference in giving (Mann-Whitney U test p=0.662). At the same time, the majority of subjects in Experiment 2 (76.15%) held prior beliefs that members of the opposing party were relatively less generous than members of their own party, and they expected ingroup members to be more generous in 57 of 99 choice pairs, on average. In this context, I define beliefs consistent with affective polarization as a guess that the ingroup member is more generous than the outgroup member in at least half of the total choice pairs, i.e., believing that their outgroup is relatively more selfish than their ingroup. Figure 2 visualizes the distribution of Democrat and Republican prior beliefs and compares them to bootstrapped realizations of relative partisan generosity.

Figure 2: Core discrepancy between prior beliefs and realized relative group generosity

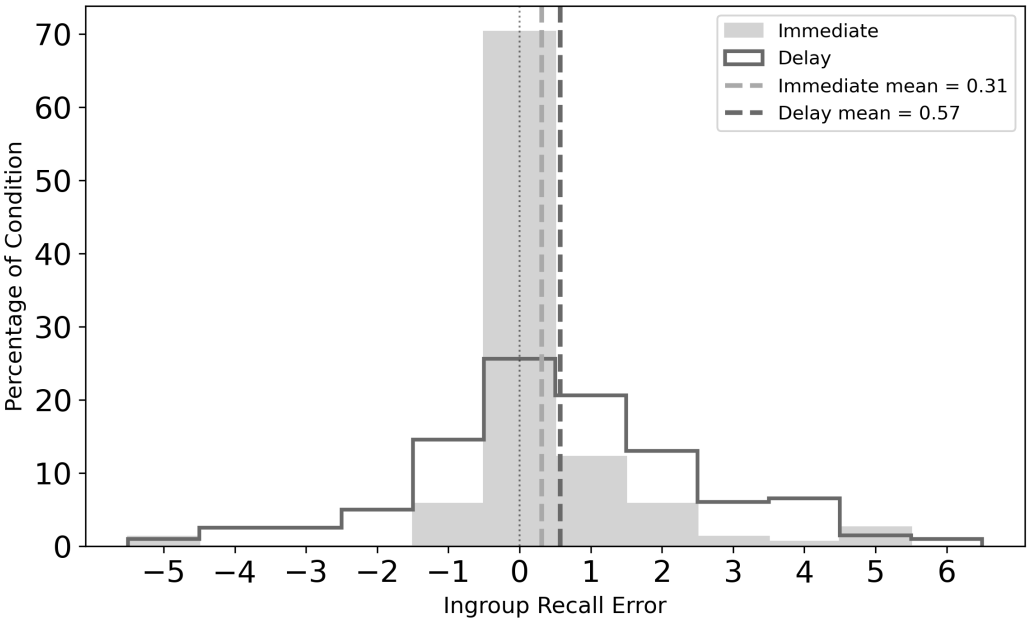


Notes: The first two panels of the figure plot the distribution of prior beliefs held by Democrats and Republicans in Experiment 2. The right side shaded in blue represents states where Democrats were more generous than Republicans in over half of choice pairs, and the left side shaded in red represents states where Republicans were more generous than Democrats in over half of choice pairs. The last panel plots the results of a bootstrap simulation (n = 10,000) where pairs of dictator game choices (one Democrat, one Republican) were randomly sampled and compared with replacement 99 times from Experiment 1 data.

3.2 Evidence of Recall Bias

In this section, I examine how subjects recall the signal about relative group generosity, which is the main outcome of interest. In the first set of results, I document a small but significant ingroup recall bias immediately after observing the signal, which becomes more pronounced after one week.

Figure 3: Distribution of recall errors by condition



Notes: This figure plots the distribution of recall errors in the *immediate* and *delay* conditions. The ingroup recall error is computed at the individual level as the number of positive pairs recalled - actual positive pairs seen.

Result 1: Immediately after observing the signal, subjects exhibit a small recall bias in favor of their group.

In Figure 3, individual recall errors are categorized, and the distribution within each condition is plotted. A positive ingroup recall error refers to recalling more positive pairs than those actually observed, and a negative ingroup recall error refers to recalling fewer positive pairs than those actually observed. In the *immediate* condition nearly 70% of the subjects correctly recalled the signal. However, on average, subjects recalled 0.31 more pairs favoring the ingroup than the true number. Using a one-sample t-test, the null hypothesis that the mean recall bias was zero is rejected ($t = 3.13, p = 0.002$). Thus, there is a small but significant overall ingroup bias in interpretation of the signal even in the *immediate* condition.[8]

Result 2: After a one week delay, the ingroup recall bias is significantly larger.

Next, I investigate recall after the one week delay. Figure 3 confirms that accurate memory is much less frequent in the *delay* condition, with less than 30% of subjects correctly recalling the number of positive pairs seen. One week after observing information on

[8]Selective attention is another potential mechanism for the ingroup recall bias in the *immediate* condition. However, this mechanism is unlikely in this setting since there is no evidence of differences in time spent looking at the signal based on its valence, as can be seen in Appendix Figure A1.

relative partisan generosity, subjects recalled 0.57 more pairs that favor the ingroup than the true number, on average. Using a one-tailed Mann-Whitney U test as preregistered, there is evidence for the alternative hypothesis that the recall bias is greater in the *delay* condition than in the *immediate* condition ($p = 0.030$).

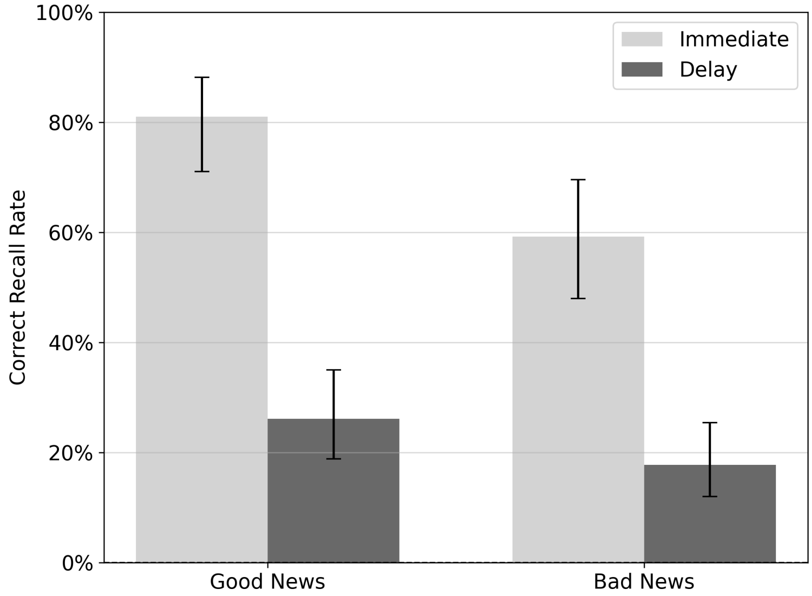
3.3 Mechanism of Recall Bias

To investigate whether the recall bias is motivated, the analysis exploits the exogenous variation in the valence of the signal, which varies the threat to partisan identity and thus the need to engage in motivated reasoning. Specifically, I consider whether subjects who received favorable signals (*good news* condition) and unfavorable signals (*bad news* condition) about the relative generosity of their group have systematically different recall dynamics. To more formally test for differential delay effects by signal valence, I estimate equations in the following form:

$$Y_i = \beta_0 + \beta_1 Delay_i + \beta_2 BadNews_i + \beta_3 I_i + \delta_0 Prior_i + \delta_1 PI_i + \gamma X_i + \epsilon_i \qquad (1)$$

where Y_i is the outcome of interest, which in this case is the recalled signal. $Delay_i$ is the dummy for being in the *delay* condition, $BadNews_i$ is a dummy for receiving *bad news*, and I_i is the interaction term of the two dummies equal to 1 if subjects received *bad news* and were in the *delay* condition. I include specifications with and without prior beliefs $Prior_i$, the interaction term of prior beliefs with the delay dummy PI_i, and a set of other controls X_i which include demographic characteristics.

Figure 4: Correct recall rate by condition

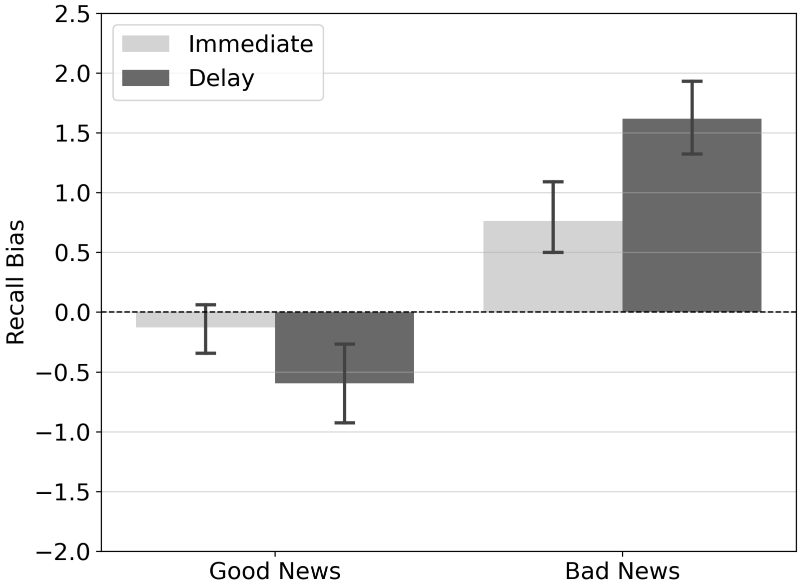


Notes: This figure plots the correct recall rate by condition and by the valence of the signal received. 95 percent confidence intervals computed using Wilson's score method.

Mechanism for Result 1 (immediate recall bias): Unfavorable signals are less likely to be correctly interpreted, leading to a systematic distortion immediately.

The differential interpretation of the signal based on its valence accounts for the initial recall bias in the *immediate* condition. Figure 4 shows the correct recall rates for each experimental condition. Upon receiving the signal, in the *immediate* condition, 81% of the subjects correctly interpreted and recalled good news, while only 59% did so for bad news ($z = 2.97, p = 0.002$). How do inaccurate subjects interpret the signal of relative partisan generosity? Figure 5 shows the average ingroup recall bias for each condition. In the *immediate* condition, the overall recall bias of subjects who received good news was not significantly different from zero. However, there was a significant ingroup bias in recall for the bad news subsample, which means unfavorable signals were interpreted with an immediate ingroup bias. The asymmetric response to good and bad signals suggests that motivated reasoning plays a role even at the early stages of information processing.

Figure 5: Average recall bias by condition



Notes: This figure plots the average recall bias by condition and by the valence of the signal received. Includes 95 percent confidence intervals.

Table 1: Recall Analysis

	Recall Bias		Recalled Signal	
	(1)	(2)	(3)	(4)
Delay	-0.469**	-0.123	-0.503**	0.132
	(0.203)	(0.676)	(0.213)	(0.612)
Bad News	0.890***	0.956***	-2.018***	-1.948***
	(0.185)	(0.186)	(0.220)	(0.217)
Delay × Bad News	1.325***	1.242***	1.225***	1.128***
	(0.299)	(0.295)	(0.313)	(0.306)
Prior		0.020**		0.026***
		(0.008)		(0.007)
Delay × Prior		-0.004		-0.008
		(0.011)		(0.010)
Constant	-0.127	-1.546***	4.886***	2.876***
	(0.108)	(0.514)	(0.141)	(0.470)
Controls	No	Yes	No	Yes
Observations	354	354	354	354
R-squared	0.273	0.299	0.196	0.239

Notes: OLS estimates of outcome *recall bias* (Columns 1 and 2) which is the recalled signal minus the observed signal and outcome *recalled signal* (Columns 3 and 4). Controls include: age, gender, political affiliation and strength of identification. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Mechanism for Result 2 (increased recall bias after delay): Unfavorable signals are further distorted in memory and recalled significantly more posi-

tively after one week.
The drop in correct recall rates in the *delay* condition confirms that the one week delay successfully led to overall memory decay. Next, I investigate how the valence of the signal influenced recall one week later. Figure 5 shows that the delay led to a negative distortion of good news to some extent and a significant positive distortion of bad news. Controlling for differences in prior beliefs across conditions, the effect of the delay is no longer statistically significant for recall errors or recall of good news, as can be seen from the coefficient on *Delay* in Columns 2 and 4 of Table 1, respectively. This suggests that imperfect memory of good news does not lead to significant distortions in any specific direction. The differential effect of the delay on recall of bad news is captured by the coefficient on the interaction term *Delay* $\times$ *Bad News*, showing that the positive distortion of bad news was significantly more pronounced after the delay. This asymmetry is inconsistent with imperfect memory in the form of noise, or attenuation of the signal strength. One explanation that could account for the asymmetry is that subjects who forgot the signal attempted to construct a guess based on their priors, which were already favorable to their group. If this was the case, as memory of the signal fades over time we would expect that favorable priors would more strongly predict favorable recall. However, the coefficient on the interaction term *Delay* $\times$ *Prior* in Column 4 is not significantly positive, and it even has the wrong sign, suggesting that subjects in the *delay* condition rely on their prior less when attempting to recall than subjects in the *immediate* condition. Thus, the valence asymmetry in recall dynamics is not driven by imperfect memory with biased priors, but rather motivated memory.

3.4 Consequences for Belief Formation

Having established that the experimental delay induces ingroup-favoring recall bias for subjects who received unfavorable signals, I now examine whether this has consequences for belief polarization at both the aggregate and individual levels.

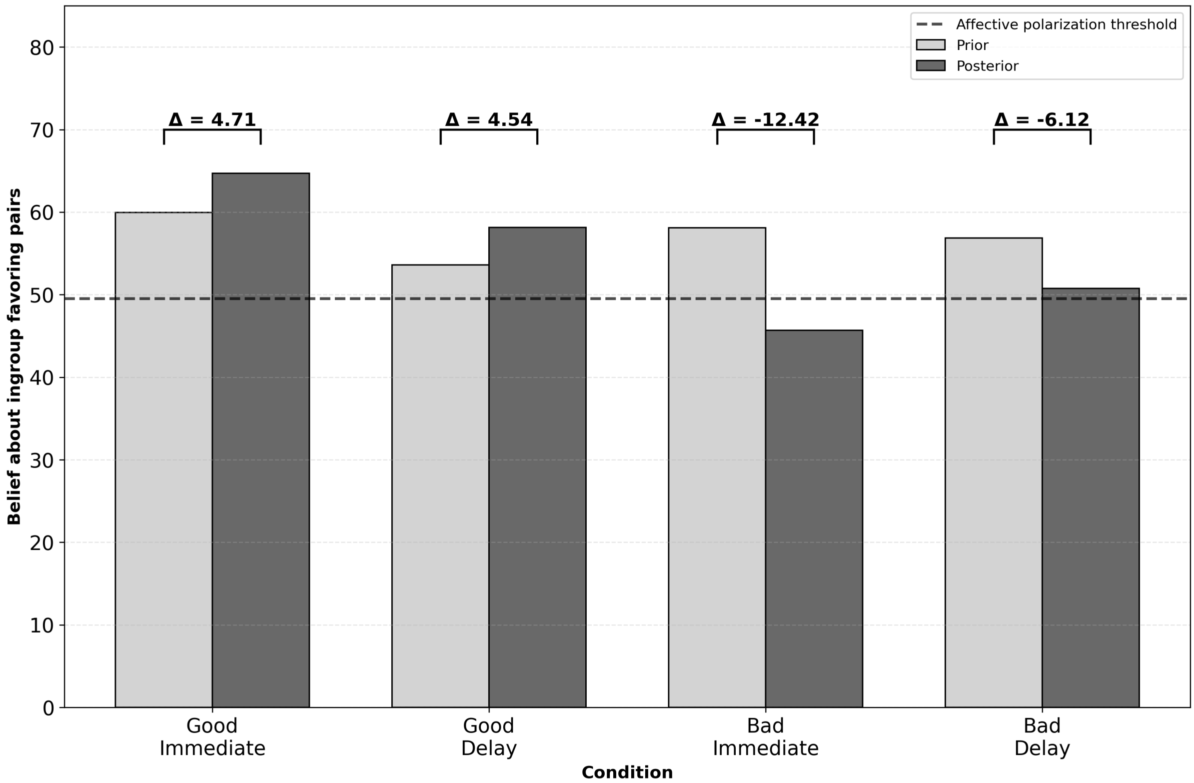
3.4.1 Persistent affective polarization at aggregate level

Figure 6 confirms that the valence of the signal shifted beliefs in the right direction in the *immediate* condition[9], and as a result, subjects who received bad news about their ingroup no longer exhibited affective polarization on average (posterior = 45.70). After the delay, however, the average posterior beliefs for those who received bad news returned to being above the threshold of affective polarization (posterior = 50.74). Belief updates were significantly attenuated after the delay ($t = -2.34, p = 0.020$). For good news,

[9]When prior beliefs already favor the ingroup, asymmetric updating is expected. Mechanically, high priors create a ceiling effect since there is less room for positive updates. Psychologically, the higher the prior, the less subjectively positive any signal becomes.

however, belief updates showed no significant differences after the delay ($t = 0.06, p = 0.953$). The differential effect that the delay has on posterior beliefs and likelihood of polarized beliefs after receiving bad news is confirmed using regression analysis, presented in Appendix Table A6. The aggregate belief dynamics are consistent with the hypothesis that positive distortion of unfavorable news over time contributes to the persistence of affective polarization in beliefs.

Figure 6: Belief dynamics by condition



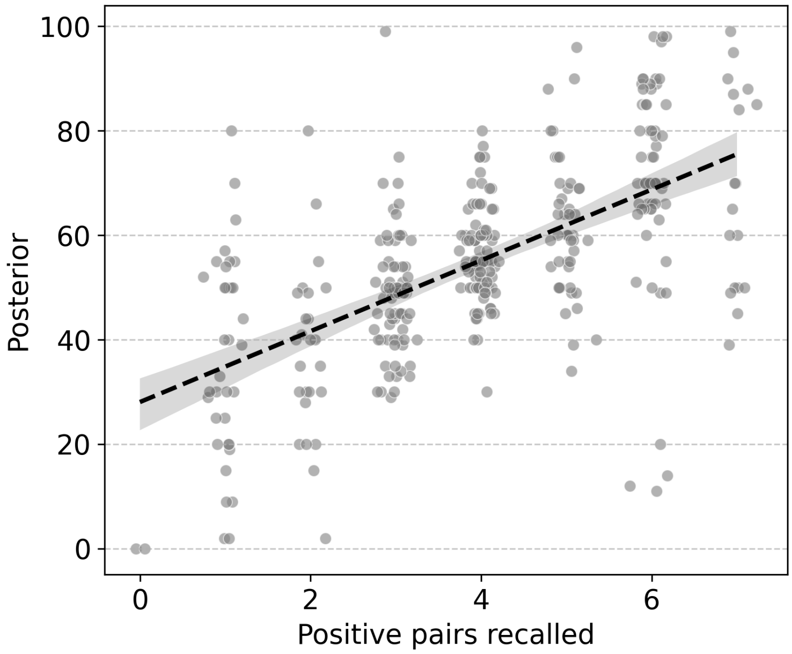
Notes: This figure plots the average prior and posterior beliefs, and the magnitude of the corresponding belief update, for each condition in Experiment 2.

3.4.2 Link between recall and belief formation at the individual level

The aggregate patterns shown in Figure 6 suggest that motivated memory distortions of bad news may sustain affective polarization in the delay condition. To investigate this mechanism, I examine the individual-level relationship between recall and belief formation. I first establish that the recalled signal is systematically related to posterior beliefs. As shown in Figure 7, there is a strong positive correlation between the recalled signal and posterior belief regarding relative partisan generosity (Spearman; $\rho = 0.61, p < 0.001$).[10]

[10]The x-values are jittered slightly to improve visual interpretation. The process involves adding a random small perturbation with mean zero to each value.

Figure 7: Relationship between recall of signal and posterior



Next, I extend the analysis to consider the role of memory errors in the belief formation process. I estimate the following equation:

$$Y_i = \alpha_0 + \alpha_1 RecallBias_i + \alpha_2 Signal_i + \alpha_3 Prior_i + \lambda X_i + \eta_i \tag{2}$$

where Y_i denotes either the posterior belief of individual i, or a binary indicator equal to 1 if the posterior belief of individual i reflects affective polarization and 0 otherwise. $RecallBias_i$ equals the recalled signal minus the observed signal, $Signal_i$. $Prior_i$ is the prior belief, and X_i is a set of controls that include age, gender, partisan identity and its strength.

Result 3: Subjects with an ingroup recall bias form beliefs about relative group generosity that are more favorable to their ingroup and are more likely to be affectively polarized.

Columns 1 and 4 of Table 2 show that holding the signal fixed, those who exhibited an ingroup recall error believed that their ingroup was relatively more generous than the outgroup and were more likely to be affectively polarized. However, since recall is endogenous, the concern in a causal interpretation is the possibility of omitted variable bias stemming from unobserved factors. One notable example is general group feelings, or the overall attachment an individual feels toward their group, which may influence both how they recall the signal and how they form their beliefs. Specifically, individuals with stronger group feelings might be predisposed to recall the signal in an ingroup-favoring way and develop beliefs more favorable to the ingroup. To mitigate this potential bias, I

Table 2: Beliefs as a function of recall bias

	Posterior Beliefs			Affective Polarization		
	(1) OLS	(2) OLS	(3) 2SLS	(4) OLS	(5) OLS	(6) 2SLS
Panel A: Bad News						
recall bias	5.389***	5.281***	8.031***	0.107***	0.105***	0.223***
	(1.024)	(0.977)	(2.679)	(0.023)	(0.023)	(0.081)
signal	7.067***	6.283***	7.628***	0.153***	0.137***	0.194***
	(1.302)	(1.323)	(2.001)	(0.037)	(0.037)	(0.057)
prior		0.337***	0.319***		0.007***	0.007***
		(0.097)	(0.100)		(0.002)	(0.003)
constant	27.468***	7.406	3.105	0.080	-0.435***	-0.620***
	(3.779)	(5.240)	(6.790)	(0.096)	(0.142)	(0.195)
controls	No	Yes	Yes	No	Yes	Yes
1st Stage F			16.869			16.869
N	181	181	181	181	181	181
R^2	0.263	0.339	0.289	0.144	0.225	0.106
Panel B: Good News						
recall bias	6.595***	5.783***		0.088***	0.081***	
	(0.853)	(0.935)		(0.026)	(0.028)	
signal	7.770***	7.484***		0.085***	0.081***	
	(1.170)	(1.162)		(0.027)	(0.028)	
prior		0.200*			0.002	
		(0.112)			(0.002)	
constant	24.881***	16.622**		0.423***	0.512**	
	(5.448)	(6.772)		(0.148)	(0.210)	
controls	No	Yes		No	Yes	
N	173	173		173	173	
R^2	0.359	0.396		0.106	0.138	

Notes: Columns (1) and (2) report OLS estimates with the outcome *Posterior Beliefs*. Columns (4) and (5) report OLS estimates of linear probability model with binary outcome *Affective Polarization*, which is 1 if posterior beliefs are greater than or equal to 50. Columns (3) and (6) report 2SLS estimates with the previous outcomes for only the bad news subsample. *prior* refers to the prior belief of the individual. Controls include age, gender, partisan identity and its strength. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

include specifications in Columns 2 and 5 that control for prior beliefs and self-reported strength of group identification, as well as other demographic characteristics. The OLS estimates are robust to these controls. The estimates indicate that for each additional favorable pair recalled beyond the true number observed, posterior beliefs are higher by approximately 5 to 6 pairs and the probability of exhibiting affective polarization is higher by 8 to 11 percentage points.

To further strengthen the causal interpretation of my estimates, I exploit the random assignment to the *delay* condition as an instrumental variable for exhibiting an ingroup bias in recall.[11] This approach only applies to the bad news subsample, since it provides a strong first-stage relationship between the *delay* condition and recall bias, while the good news subsample does not. In particular, I instrument for recall bias with being assigned to the *delay* condition and estimate the following equations:

$$Y_i = \alpha_0 + \alpha_1 \widehat{RecallBias}_i + \alpha_2 Signal_i + \alpha_3 Prior_i + \lambda X_i + \eta_i \tag{3}$$

$$RecallBias_i = \beta_0 + \beta_1 Delay_i + \beta_2 Signal_i + \beta_3 Prior_i + \gamma X_i + \epsilon_i \tag{4}$$

where Equation (4) is the first-stage regression and Equation (3) is the second-stage regression.

Result 4: Exhibiting an ingroup recall bias significantly increases the ingroup bias in posterior beliefs and the probability of affective polarization. The two-stage least squares estimates support a causal interpretation of the relationship between recall and belief formation. Exploiting the variation in recall induced by the delay, I find that a one-unit increase in one's ingroup recall error increases posterior beliefs by approximately 8 ingroup favoring pairs out of 99 (Column 3 of Panel A in Table 2) and increases the probability of exhibiting affective polarization in beliefs by 22 percentage points (Column 6, Panel A, Table 2). These individual-level results confirm that the positive distortion of unfavorable news over time influences the belief formation process and contributes to the persistence of affective polarization that we observe in the aggregate belief dynamics.

3.5 Consequences for Intergroup Behavior

3.5.1 Dynamics of intergroup behavior

Having established that biased recall of bad news sustains affective polarization in beliefs, I now examine whether these distorted beliefs translate into discriminatory intergroup behavior. At the end of Experiment 2, subjects made two intergroup choices: they indicated their willingness to pay (WTP) to match with an ingroup versus outgroup dictator to receive an allocation from, and they decided whether they wanted to reduce an ingroup or outgroup member's payoff at no cost to themselves.

The valence of the signal had immediate effects on intergroup behavior. Subjects receiving bad news about their group's relative generosity were substantially less likely to favor their ingroup: only 64.47% were willing to pay to match with an ingroup dictator, compared to 83.54% of those receiving good news (Two-proportion Z-test; $z = 2.71, p = 0.007$). Similarly, bad news increased ingroup punishment, with 26.32% of subjects choosing to reduce an ingroup member's payoff versus 12.66% choosing to do

[11] I assess the plausibility of the identifying assumptions for this IV strategy in Appendix A.4.

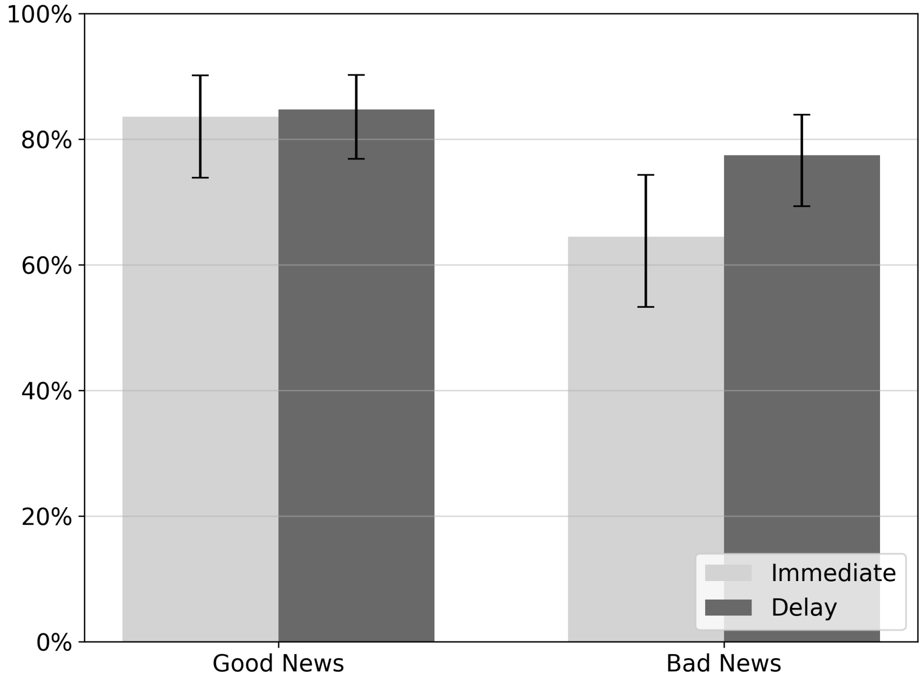
so in the good news condition ($z = -2.15, p = 0.031$). Figures 8 and 9 illustrate these patterns across experimental conditions.

After the one week delay, however, behavioral patterns shifted dramatically, but only for subjects who received bad news. Among this group, the share of subjects willing to pay for matching with an ingroup dictator rebounded from 64.47% to 77.42% ($z = -1.99, p = 0.047$), while the share punishing the ingroup dropped sharply from 26.32% to 11.29% ($z = 2.75, p = 0.006$). By contrast, the delay produced no significant changes among subjects who received good news. The share of subjects willing to pay for ingroup matching remained stable (immediate: 83.54%, delay: 84.68%; $z = -0.21, p = 0.832$), as did the share punishing the ingroup (immediate: 12.66%, delay: 15.32%; $z = -0.52, p = 0.605$). The more formal analysis for a differential effect of the delay is presented in Appendix Table A7.[12]

This asymmetry in behavioral responses mirrors the corresponding asymmetries by signal valence in recall and belief formation. The results suggest that as memory of the unfavorable signal about one's group image is distorted over time and is no longer accurately reflected in beliefs, discrimination toward the outgroup increases and accountability of ingroup members decreases.

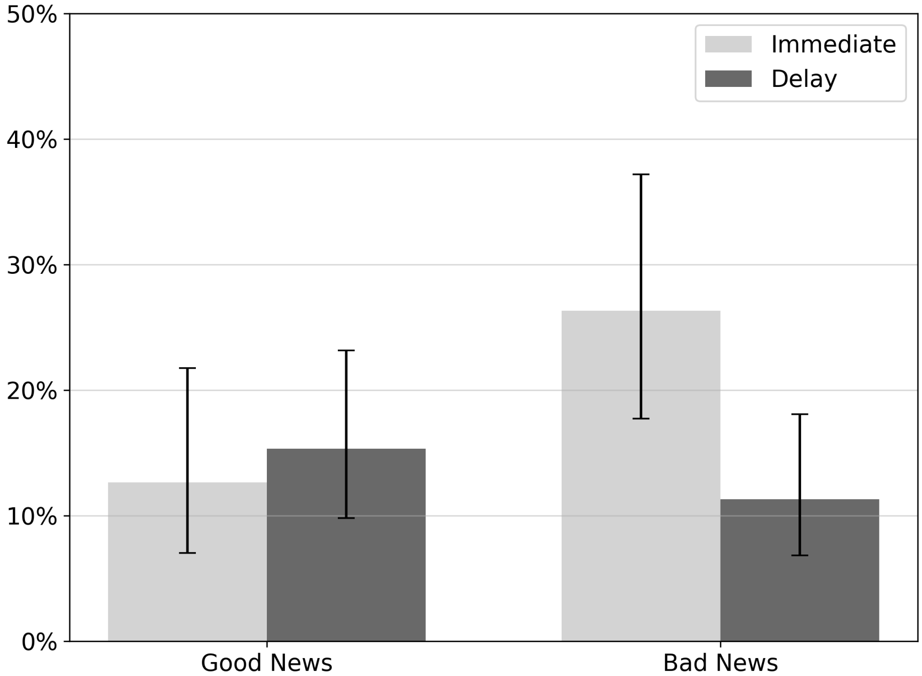
[12] I cannot conclude from the formal analysis that there is a statistically significant differential effect of delay across news valence for exhibiting a positive willingness to pay, as the difference-in-differences regression does not yield significant results. However, the z-tests of proportions do establish that the delay has a statistically significant effect for bad news but not for good news. The DID regression provides a more stringent test of whether the delay effect differs across news valence by formally testing the interaction term, whereas the separate z-tests only establish whether delay has an effect within each signal type.

Figure 8: Share of subjects willing to pay for ingroup dictator



Notes: This figure plots share of subjects willing to pay a positive amount to match with an ingroup rather than outgroup dictator, by experimental condition. 95 percent confidence intervals computed using Wilson's score method.

Figure 9: Share of subjects reducing ingroup payoffs

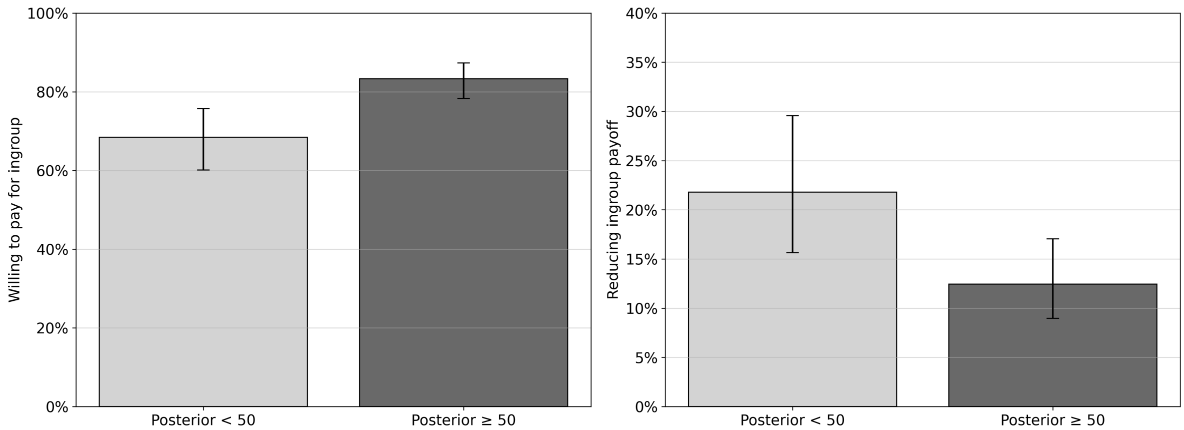


Notes: This figure plots the share of subjects deciding to reduce a random ingroup member's payoff, by experimental condition. 95 percent confidence intervals computed using Wilson's score method.

3.5.2 Link between beliefs and intergroup behavior

The aggregate patterns raise the question of whether individual-level beliefs formed through motivated memory shape these intergroup behaviors. To investigate this, I examine how posterior beliefs about relative group generosity relate to the two intergroup choices. For the WTP task, beliefs about relative group generosity are directly relevant to subjects' payoffs, as they should be willing to pay a premium if they expect to receive more from an ingroup than from an outgroup dictator. At the same time, if Democratic and Republican dictators give equal amounts to their partners on average, a positive WTP represents an expected welfare loss. For the payoff reduction task, beliefs about relative group generosity are not relevant to one's own monetary payoff. However, this choice presents an opportunity to punish selfish behavior. Figure 10 confirms that subjects with ingroup-favoring posterior beliefs are more likely to be willing to pay for an ingroup dictator ($z = -3.37, p < 0.001$) and less likely to reduce ingroup payoffs ($z = 2.41, p = 0.016$). In other words, believing that the ingroup is more generous than the outgroup is associated with greater outgroup discrimination and less ingroup accountability. These individual-level results confirm that the distorted, polarized beliefs documented earlier are associated with discriminatory intergroup behavior. These results tie together downstream effects of motivated memory, demonstrating the consequences for belief formation and subsequent behavior.

Figure 10: Relationship between posterior beliefs and intergroup behavior



Notes: This figure shows the differences in intergroup behavior by subject whose beliefs exhibit affective polarization (posterior beliefs ≥ 50) and those who do not (posterior beliefs < 50). The left panel shows shares of subjects willing to pay to match with an ingroup rather than outgroup dictator, and the right panel shows shares of subjects deciding to reduce a random ingroup member's payoff. 95 percent confidence intervals computed using Wilson's score method.

3.6 Discussion: Effect of early recall/belief formation

As mentioned previously, participants in the *immediate* condition returned for Part 2 of the experiment to complete the recall task for a second time, allowing for a within-subject comparison of the recall bias after the one week delay. These subjects exhibited an average recall bias of 0.34, compared to their average recall bias from the previous week, 0.31. According to a one-sided Wilcoxon signed-rank test, we cannot reject that the recall bias remained the same ($p = 0.25$). Part 2 recall for *immediate* condition subjects occurred after the same amount of time had passed for *delay* subjects, but these conditions differed in two potentially important ways. *Immediate* subjects were asked to recall the signal immediately after observing it, and they were also asked to form the beliefs moments later. This suggests that early recall or belief formation may limit the distorting capability of motivated memory. Possible explanations from the psychology literature include the effect of signal "rehearsal" or the recall task serving as a situational cue linking the two parts of the experiment. Another plausible explanation could be that beliefs about the state of the world are more memorable or readily available than the signal itself. In this case when prompted to recall the signal, instead of retrieving the signal itself, *immediate* condition subjects who updated their beliefs in Part 1 may use them to back out the signal. This study cannot speak to which potential explanation is driving the result. Rather, it provides suggestive evidence that rehearsal and/or early belief formation may mitigate the motivated memory and its consequences.

Table 3: Part 1 vs Part 2 recall bias

	Part 1 Mean	Part 2 Mean	Mann-Whitney U test / Signed-rank test
Between Subject	0.31	0.57	$p = 0.03$
Within Subject	0.31	0.34	$p = 0.25$

The table reports the between subject comparison of the mean recall bias in Part 1 for the *immediate* condition, and in Part 2 for the *delay* condition, as well as the within subject comparison of the mean recall bias in Part 1 and Part 2 for the *immediate* condition. It also reports one-tailed non-parametric tests of the null hypothesis that the recall bias did not increase between Part 1 and Part 2.

4 Conclusion

In this paper, I investigate whether motivated memory distortions serve as a key psychological mechanism sustaining affective polarization. Through two complementary online experiments, I examine how partisan identity influences the recall of signals informative of relative group generosity and trace the consequences of these memory biases for belief formation and intergroup behavior.

My findings provide novel evidence that motivated memory plays a crucial role in sustaining partisan animosity in beliefs. I confirm a substantial belief-behavior discrepancy, with a majority of partisans expecting their ingroup to be more generous than the outgroup despite no differences in dictator game choices. Crucially, subjects systematically distort their recall of relative partisan generosity in favor of their own group, with this bias becoming more pronounced over time and driven primarily by misremembering unfavorable information. This differential effect underscores the role of motivated cognition, where the psychological need to defend one's group image against negative information drives memory distortion. Finally, I document that these memory distortions have direct causal effects on belief formation, and these polarized beliefs are in turn associated with costly discrimination toward outgroup members and reduced accountability toward ingroup members.

These results have implications for understanding the persistence of affective polarization and designing effective interventions. My findings help explain why corrective interventions may fail to produce lasting change in partisan attitudes. Even when people initially update their beliefs in response to contrary evidence, motivated memory gradually erodes these gains over time. This mechanism suggests that interventions targeting affective polarization may need to account for memory dynamics to achieve durable effects. However, my results also suggest a potential solution: participants who immediately recalled the signal and formed beliefs showed no evidence of further memory distortion one week later, indicating that rehearsal and/or early belief formation may counteract motivated memory. More broadly, by documenting how memory biases translate into costly discrimination and weakened ingroup accountability, this research underscores the costs of affective polarization while identifying memory dynamics as a crucial component to consider in efforts to address the issue.

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A Appendix

A.1 Summary and Balance

A.1.1 Experiment 1

Table A1: Experiment 1 summary statistics and overall balance

	Republican (N=99) Mean (SD)	*Democrat* (N=94) Mean (SD)	t-test p-value
age	45.28 (13.41)	38.32 (12.09)	0.000
male	0.49 (0.50)	0.51 (0.52)	0.832
identification	7.59 (2.03)	7.55 (2.00)	0.910
allocation	35.04 (28.36)	35.57 (26.97)	0.893

A.1.2 Experiment 2

This section presents the balance tables by timing condition for each observable characteristic and the signal received. The full sample is presented first, and the next balance tables separate the data into subsamples based on the valence of the signal. The prior is imbalanced between the *immediate* and *delay* conditions, which I control for in the regression analyses. Assessing overall balance following random assignment, an omnibus F-test of joint orthogonality with randomization inference p-values (Kerwin et al. 2024) does not reject the null of balance across all covariates.

Table A2: Experiment 2 summary statistics and overall balance

	immediate (N=155) Mean (SD)	*delay* (N=235) Mean (SD)	t-test p-value
prior	59.06 (13.97)	55.33 (14.37)	0.011
identification	0.54 (0.50)	0.59 (0.49)	0.379
male	0.50 (0.50)	0.49 (0.50)	0.886
age	43.54 (13.66)	40.90 (13.58)	0.062
democrat	0.52 (0.50)	0.46 (0.50)	0.258
positive pairs	3.59 (1.77)	3.41 (1.80)	0.345

Table A3: Good News subsample: summary statistics and balance

	immediate (N=79) Mean (SD)	*delay* (N=111) Mean (SD)	t-test p-value
prior	59.97 (13.71)	53.61 (15.82)	0.004
identification	0.56 (0.50)	0.58 (0.50)	0.790
male	0.56 (0.50)	0.53 (0.50)	0.730
age	43.82 (13.68)	41.66 (14.09)	0.290
democrat	0.54 (0.50)	0.46 (0.50)	0.252
positive pairs	5.01 (1.01)	4.99 (1.00)	0.884

Table A4: Bad News subsample: summary statistics and balance

	immediate (N=76) Mean (SD)	*delay* (N=124) Mean (SD)	t-test p-value
prior	58.12 (14.26)	56.86 (12.81)	0.531
identification	0.53 (0.50)	0.60 (0.49)	0.334
male	0.43 (0.50)	0.45 (0.50)	0.811
age	43.24 (13.73)	40.22 (13.12)	0.127
democrat	0.50 (0.50)	0.47 (0.50)	0.660
positive pairs	2.11 (1.00)	2.00 (1.00)	0.472

A.2 Attrition

Of the 440 participants recruited for Part 1, the overall attrition rate was 11.36% for a final sample size of 390 who completed Part 2 of the study. There was no evidence of differential attrition across observables or experimental manipulations.

Table A5: Attrition analysis: summary statistics and balance

	returned (N=390) Mean (SD)	*attrition* (N=50) Mean (SD)	t-test p-value
prior	56.81 (14.31)	57.64 (14.93)	0.712
identification	0.57 (0.50)	0.58 (0.50)	0.886
male	0.49 (0.50)	0.60 (0.49)	0.153
age	41.95 (13.65)	39.42 (13.75)	0.226
democrat	0.49 (0.50)	0.58 (0.50)	0.220
positive pairs	3.48 (1.79)	3.80 (1.86)	0.258
delay	0.60 (0.49)	0.64 (0.48)	0.610

A.3 Robustness: Selection

In this section, I address the concern that selection bias is introduced in the recall measure due to the presence of the option to select "I don't recall" allowing subjects to opt out of recall. In Part 1 of the experiment, no *immediate* condition subjects selected this option, but in Part 2 of the experiment, 36 *delay* condition subjects (15.32%) selected this option. Out of all observable characteristics, age is the only significant predictor of selecting "I don't recall". Subjects in the *delay* condition selecting the option were 5.82 years older on average than subjects attempting to recall. Prior beliefs, level of partisan identification, and the valence of the signal are not predictive of selecting the option. In addition, conditional on not selecting "I don't recall", age was not related to the number of favorable pairs recalled. The evidence suggests that opting out of recall is driven by age-related factors, such as imperfect memory and/or a preference for truth-telling rather than motivated memory.

Next, since participants in the *immediate* condition returned in Part 2 of the experiment to repeat the recall task, I can examine the subset of those who selected "I don't recall" after the same period of time. 12 subjects (7.74%) selected this option, and they were 4.84 years older, on average, than subjects attempting to recall. To test the robustness of Result 2 in Section 3.2, which compares the Part 1 recall bias in the *immediate* condition to the Part 2 recall bias of the *delay* condition, I can exclude subjects in the immediate condition who selected "I don't recall" when they returned in Part 2. Selection effects correlated with age, such as imperfect memory or a preference for truth-telling, should thus be held constant across conditions. I find that the increase in the ingroup recall bias after one week is robust and more significant when attempting to control for selection effects (Mann-Whitney U; $p = 0.023$, one-sided).

A.4 Identifying assumptions for IV strategy

Exclusion restriction: The key identification assumption is that the one week delay influences posterior beliefs solely through its impact on recall, with no direct effect on these outcomes. A potential violation of the exclusion restriction could arise if significant external events during the delay period influenced participants' attitudes toward political groups, thereby directly affecting how they engaged in Part 2 of the experiment. There are two reasons to doubt that this occurred.

First, Google Trends data in Figure A2 indicate that public interest in search terms related to partisan groups remained relatively stable between Part 1 (week of December 1, 2024) and Part 2 (week of December 8, 2024) of the experiment. Search interest declined by exactly one index point for all four terms (Democrats, Republicans, liberals, and conservatives) over this week, roughly half the median week-to-week change observed across the surrounding four months. The party and ideological labels moved identically, rather than diverging, suggesting that no asymmetric partisan development intervened.

Second, suppose a significant event occurred between Parts 1 and 2 of the experiment that shifted perceptions about either Democrats or Republicans. If the news shifted perceptions of both groups in a certain direction, we might expect the effect of the delay on belief updates in the ingroup direction to be larger for one group than the other. If one side avoided the news and it did not change their perceptions, we would also expect to see a differential effect of the delay on belief updates by political affiliation. Holding the signal, the recalled signal, prior beliefs, and demographic characteristics fixed, I find no such effect. The interaction between the delay and party identification is statistically insignificant for posterior beliefs ($p = 0.13$), affective polarization ($p = 0.16$), willingness to pay to match with an ingroup dictator ($p = 0.68$), and ingroup punishment ($p = 0.60$). Figure A3 plots the corresponding group means for posterior beliefs. Given this stability, it is unlikely that the delay influenced posterior beliefs and intergroup behavior through any channel other than its intended effect on signal recall.

Violation of monotonicity and identification under constant treatment effect assumption: The 2SLS estimates identify the local average treatment effect (LATE) of having a positive recall error for compliers under the assumption of monotonicity. The monotonicity condition states that the ingroup recall error would be weakly greater in the *delay* condition than the *immediate* condition for all the subjects. Because accurate memory is prevalent in the *immediate* condition while negative recall error is more common in the *delay* condition, it is plausible that some subjects would exhibit accurate recall immediately but exhibit a negative error after the delay. For these individuals, the recall error would become more negative going from the *immediate* to the *delay* condition, directly contradicting monotonicity. An alternative assumption to invoke for 2SLS to identify a treatment effect is that there is a constant treatment effect (Imbens & Angrist 1994). Although this assumption is implausible in many settings where we expect heterogeneous effects such as returns to schooling, I argue that the tight link between recall and beliefs in this context makes this a plausible assumption. Specifically, the constant treatment effect assumption requires that recalling one additional ingroup-favorable pair shifts beliefs about relative group generosity by the same amount for all subjects.

A.5 Robustness: Destruction as another measure of affective polarization

Although the joy-of-destruction game is not a traditional measure of affective polarization, it has been used as a behavioral indicator of the phenomenon by Voelkel et al. (2023), measuring a stronger form of animosity. As a measure of affective polarization supplementary to beliefs I also consider net outgroup destruction, or the amount that subjects reduced outgroup payoffs by. When subjects reduced ingroup payoffs this variable is coded as a negative integer, and when subjects took no action this variable is coded as zero. Consistent with the previous results on belief updates, subjects who received bad news about their ingroup no longer exhibited affective polarization on average as we cannot reject that net outgroup destruction is equal to zero ($t = -0.16, p = 0.874$). After the delay, however, net outgroup destruction is significantly greater than in the immediate condition ($t = -2.28, p = 0.023$).

A.6 Additional Tables and Figures

Table A6: Difference-in-Differences: Beliefs

	Posterior		Affective Polarization	
	(1)	(2)	(3)	(4)
Delay	-6.530***	-4.156*	-0.117**	-0.072
	(2.380)	(2.221)	(0.055)	(0.056)
Bad News	-18.986***	-18.522***	-0.439***	-0.425***
	(2.837)	(2.640)	(0.068)	(0.066)
Delay × Bad News	11.575***	10.083***	0.255***	0.221**
	(3.573)	(3.358)	(0.091)	(0.088)
Prior		0.315***		0.006***
		(0.087)		(0.002)
Constant	64.684***	39.476***	0.873***	0.414***
	(1.741)	(4.623)	(0.037)	(0.120)
Controls	No	Yes	No	Yes
Observations	390	390	390	390
R-squared	0.133	0.215	0.108	0.163

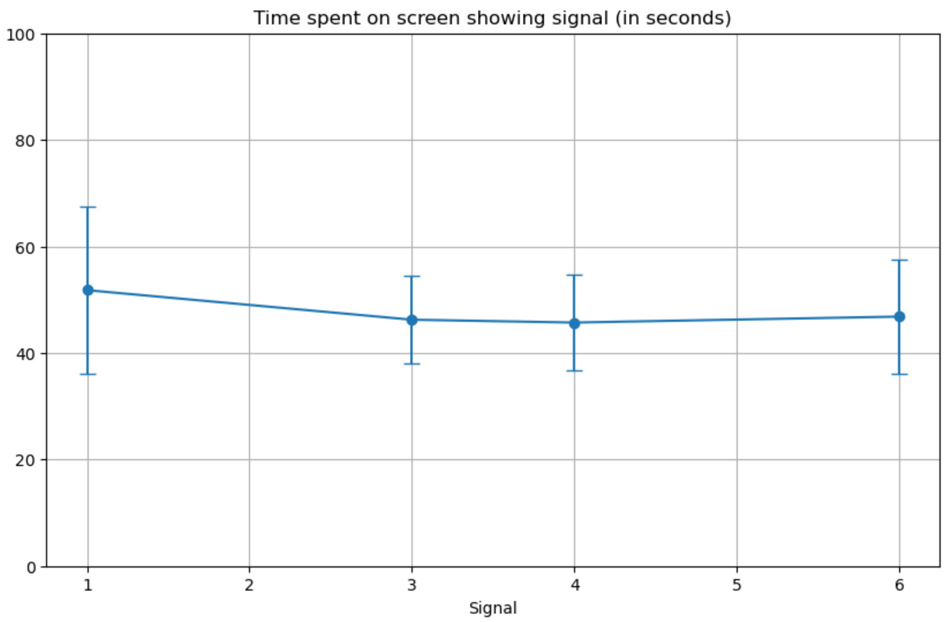
Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.
Controls include: age, gender, political affiliation and identification.

Table A7: Difference-in-Differences: Behavior

	Positive WTP		Ingroup Punishment	
	(1)	(2)	(3)	(4)
Delay	0.011	0.028	0.027	0.012
	(0.054)	(0.054)	(0.051)	(0.049)
Bad News	-0.191***	-0.193***	0.137**	0.133**
	(0.069)	(0.068)	(0.063)	(0.062)
Delay × Bad News	0.118	0.112	-0.177**	-0.171**
	(0.086)	(0.084)	(0.077)	(0.075)
Prior		0.002		-0.001
		(0.001)		(0.002)
Constant	0.835***	0.567***	0.127***	0.140
	(0.042)	(0.116)	(0.037)	(0.115)
Controls	No	Yes	No	Yes
Observations	390	390	390	390
R-squared	0.032	0.053	0.023	0.059

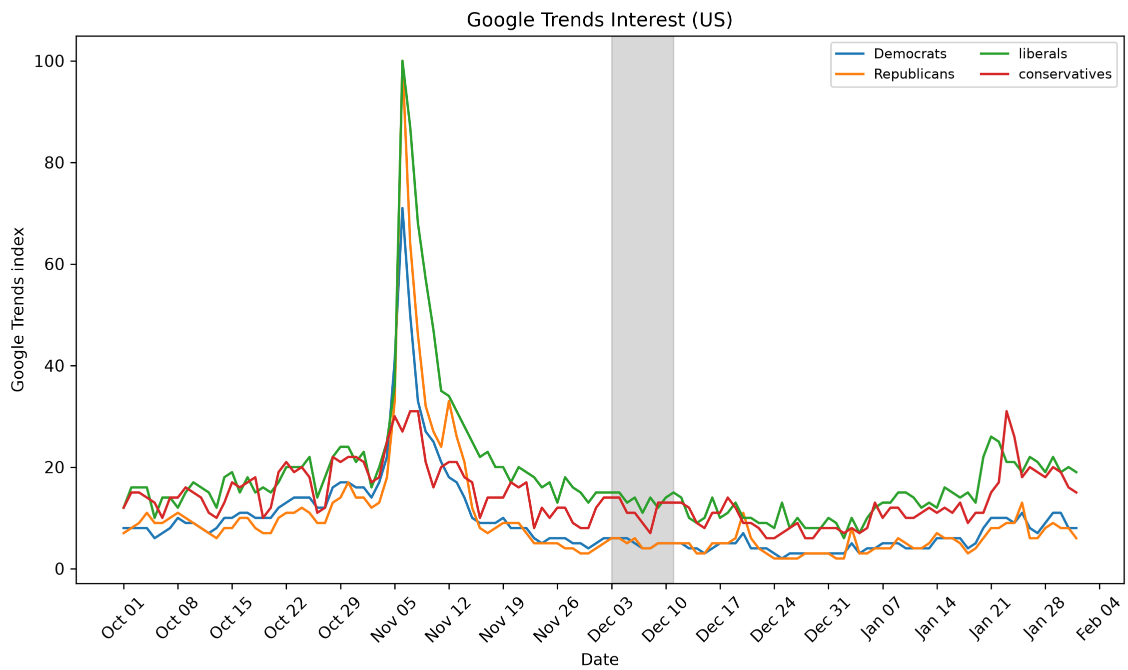
Notes: OLS estimates with robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10.
Controls include: age, gender, political affiliation and identification.

Figure A1: Attention to signal



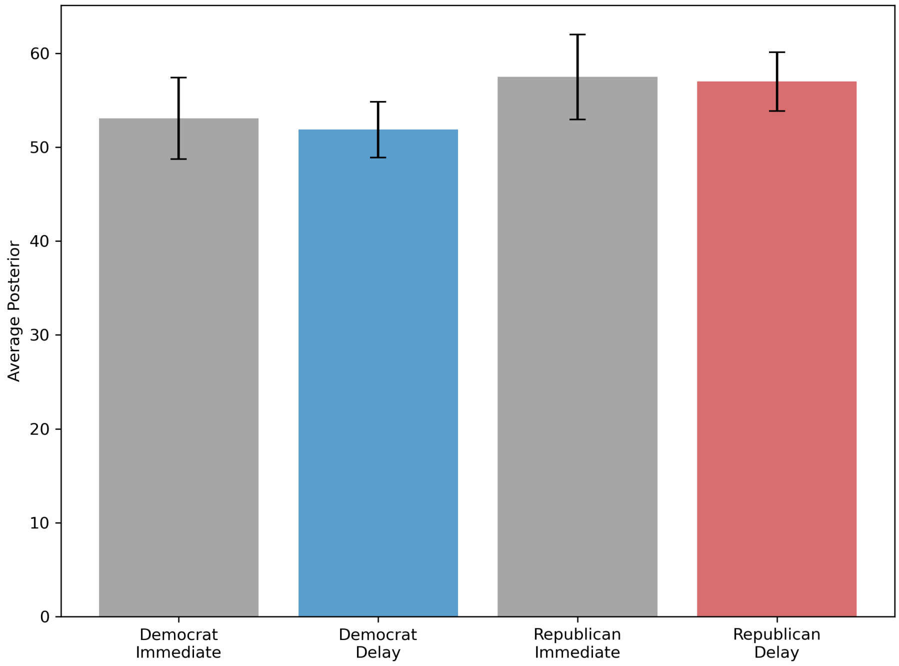
This figure shows the average number of seconds that subjects spent on the screen showing the signal in Experiment 2, by signal observed.

Figure A2: Google Trends



This figure shows fluctuations in Google Search interest for the terms "Democrats", "Republicans", "liberals", and "conservatives" in the United States. Part 1 of Experiment 2 took place on December 3rd, and Part 2 took place between December 10th and 12th.

Figure A3: Effect of delay on beliefs by political affiliation



A.7 Experiment Screenshots

Figure A4: Signal

Pair Number	Outcome	Democrat	Republican
1	In this pair, the Democrat was the 'lesser giver'.	Democrat	Republican
2	In this pair, the Republican was the 'lesser giver'.	Democrat	Republican
3	In this pair, the Democrat was the 'lesser giver'.	Democrat	Republican
4	In this pair, the Democrat was the 'lesser giver'.	Democrat	Republican
5	In this pair, the Republican was the 'lesser giver'.	Democrat	Republican
6	In this pair, the Republican was the 'lesser giver'.	Democrat	Republican
7	In this pair, the Democrat was the 'lesser giver'.	Democrat	Republican

Figure A5: Recall task

The correct number of pairs that you saw is 7. In each of the 7 pairs, you saw the result of comparing a **real decision** made by a Democrat and a Republican as to how much, out of their own 100 tokens, to send to an anonymous partner.

Out of the 7 pairs of Individual A's decisions, in how many was the Republican the 'lesser giver'—i.e., they gave less to their partner and kept more for themselves than the Democrat?

BONUS: You will earn a bonus of 100 tokens if you correctly recall the true number of pairs where the Republican was the 'lesser giver' than the Democrat.

- ○ 0
- ○ 1
- ○ 2
- ○ 3
- ○ 4
- ○ 5
- ○ 6
- ○ 7
- ○ I don't recall

Submit

Figure A6: Belief elicitation

From all participants in the previous section of this study, the computer randomly generated 99 pairs of decisions, each consisting of one Democrat and one Republican in the role of Individual A in the Giving Game. **In each pair, the participant who kept more for themselves and gave less to their partner was labeled as the 'lesser giver.'**

You will guess the number of pairs where the Republican was the 'lesser giver'—meaning they kept more for themselves and gave less to their partner than the Democrat did—and the number of pairs where the Democrat was the 'lesser giver'. In each of the 99 pairs, one "Individual A" is labeled as the 'lesser giver' (if they made the exact same decision, the computer randomly selects one of them as the 'lesser giver').

BONUS: If your guess is within two pairs of being correct, you will earn an additional 100 tokens.

Out of 99, how many times do you think the Republican was the 'lesser giver'?

Out of 99, how many times do you think the Democrat was the 'lesser giver'?

How confident are you that your guess is within two pairs of being correct?

- ○ Very unconfident
- ○ Somewhat unconfident
- ○ Somewhat confident
- ○ Very confident

Submit

Figure A7: Multiple Price List

You will receive the tokens from the previous giving decision of one Individual A in the Giving Game after making a series of choices. **For each row below, you must choose whether you prefer Option 1 or Option 2**. (Selecting Option 1 will automatically fill all rows above with Option 1, and selecting Option 2 will fill all rows below with Option 2):

- **Option 1**: Match with a Republican and receive tokens based on their Individual A decision.
- **Option 2**: Match with a Democrat and receive tokens based on their Individual A decision, **with an additional token adjustment (Either pay or receive tokens for choosing Option 2)**.

The computer will randomly select one decision/row to count for payment.

- **If you prefer Option 1 in the selected row**, you'll receive the number of tokens sent by a randomly chosen Republican, multiplied by three.
- **If you prefer Option 2 in the selected row**, you'll receive the number of tokens sent by a randomly chosen Democrat, multiplied by three, **plus or minus the token adjustment.**

Example: Suppose row 16 is randomly selected by the computer, and in that row you prefer Option 2.

- You will be matched with a randomly selected Democrat, who sent X tokens to their anonymous partner, resulting in $3 \cdot X$ tokens for you.
- You also receive 50 tokens for choosing Option 2.
- **Your Earnings**: $3 \cdot X + 50$ tokens

Decision	Option 1: Match with a Republican	OR	Option 2: Match with a Democrat
1	No Token Adjustment	1 ○ ○ 2	Pay 100 tokens
2	No Token Adjustment	1 ○ ○ 2	Pay 90 tokens
3	No Token Adjustment	1 ○ ○ 2	Pay 80 tokens
4	No Token Adjustment	1 ○ ○ 2	Pay 70 tokens
5	No Token Adjustment	1 ○ ○ 2	Pay 60 tokens
6	No Token Adjustment	1 ○ ○ 2	Pay 50 tokens
7	No Token Adjustment	1 ○ ○ 2	Pay 40 tokens
8	No Token Adjustment	1 ○ ○ 2	Pay 30 tokens
9	No Token Adjustment	1 ○ ○ 2	Pay 20 tokens
10	No Token Adjustment	1 ○ ○ 2	Pay 10 tokens
11	No Token Adjustment	1 ○ ○ 2	No Token Adjustment
12	No Token Adjustment	1 ○ ○ 2	Receive 10 tokens
13	No Token Adjustment	1 ○ ○ 2	Receive 20 tokens
14	No Token Adjustment	1 ○ ○ 2	Receive 30 tokens
15	No Token Adjustment	1 ○ ○ 2	Receive 40 tokens
16	No Token Adjustment	1 ○ ○ 2	Receive 50 tokens
17	No Token Adjustment	1 ○ ○ 2	Receive 60 tokens
18	No Token Adjustment	1 ○ ○ 2	Receive 70 tokens
19	No Token Adjustment	1 ○ ○ 2	Receive 80 tokens
20	No Token Adjustment	1 ○ ○ 2	Receive 90 tokens
21	No Token Adjustment	1 ○ ○ 2	Receive 100 tokens

Please click Submit when you have made your decision.

Submit

Figure A8: Joy of Destruction game

Earlier in this study, all participants, including you, earned 100 tokens by observing 7 pairs of Individual A decisions. Now, **you have the opportunity to reduce up to 100 tokens from another participant's earnings**—and another participant can reduce up to 100 tokens of your earnings.

You may choose to reduce the earnings of:

- An anonymous Democrat
- An anonymous Republican
- No one

Notes:

- Your choice is completely **anonymous**. The participant whose earnings you might reduce will not know your identity or any personal information about you. They will be informed that an anonymous participant chose to reduce their earnings and by how much, but no further details will be provided.
- **Your choice can reduce the earnings of one participant, at most**. If you decide to reduce the earnings of a specific group, your choice will apply to a randomly selected member of that group, provided there are eligible participants currently taking the study.
  - Your choice cannot reduce your own earnings.
- **Your earnings can be reduced by one participant, at most**. Another participant may have chosen to reduce the earnings of someone in your group, and if you are selected out of all eligible participants, their choice will affect your earnings.
  - The participant whose earnings you may choose to reduce may or may not be the same participant who may choose to reduce your earnings.
- Your choice must be a multiple of 10 tokens (e.g., 0, 10, 20, ..., 100).

Please indicate your choice below.

Democrat Republican Neither

Reduction choice: 0 tokens

Submit